In [22]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("drug200.csv")
```

In [23]:
```
# 1. Shape of dataset
print("Shape of dataset:")
print(df.shape)
```

```
Shape of dataset:
(200, 6)
```

In [24]:
```
# 2. Column names
print("\nColumns:")
print(df.columns)
```

```
Columns:
Index(['Age', 'Sex', 'BP', 'Cholesterol', 'Na_to_K', 'Drug'], dtype='object')
```

In [25]:
```
# 3. Data types
print("\nData Types:")
print(df.dtypes)
```

```
Data Types:
Age              int64
Sex             object
BP              object
Cholesterol     object
Na_to_K        float64
Drug            object
dtype: object
```

In [26]:
```
# 4. Missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Age            0
Sex            0
BP             0
Cholesterol    0
Na_to_K        0
Drug           0
dtype: int64
```

In [27]:
```
# Total missing values
print("\nTotal Missing Values:", df.isnull().sum().sum())
```

```
Total Missing Values: 0
```

In [28]:
```
# 5. Duplicate values
print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

```
Duplicate Rows:
0
```

In [29]:
```
# 6. Statistical summary for numerical columns
print("\nDescribe Numerical Data:")
```

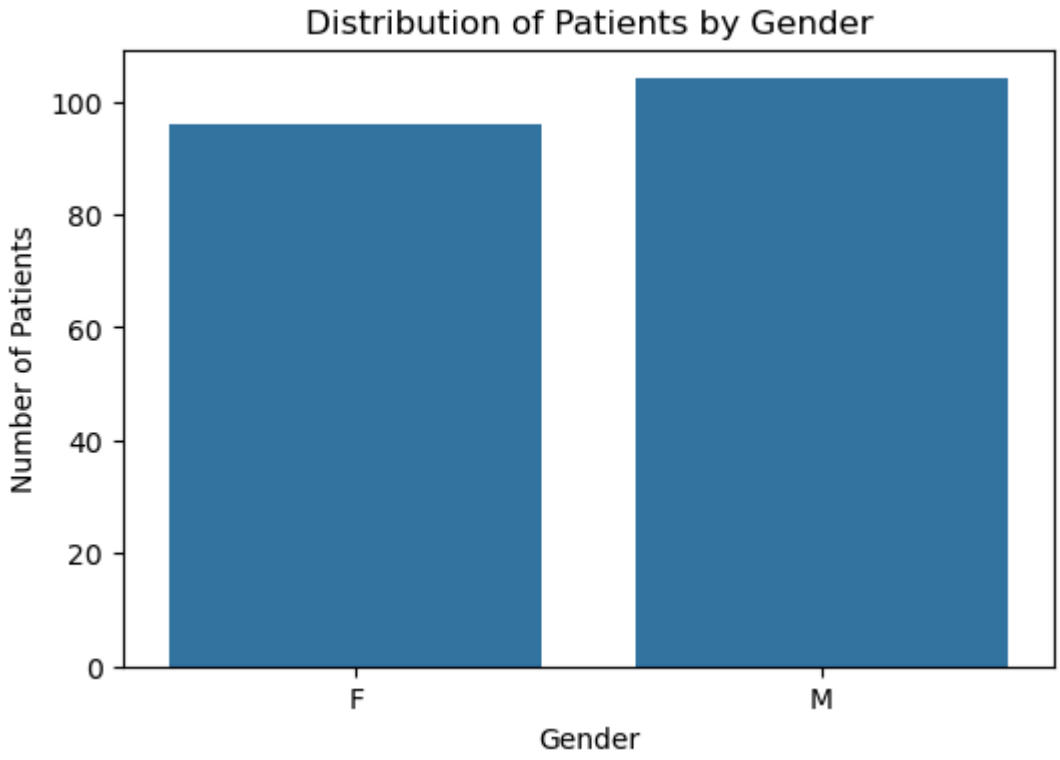
```
print(df.describe())
```

```
Describe Numerical Data:
              Age      Na_to_K
count  200.000000  200.000000
mean    44.315000   16.084485
std     16.544315    7.223956
min     15.000000    6.269000
25%     31.000000   10.445500
50%     45.000000   13.936500
75%     58.000000   19.380000
max     74.000000   38.247000
```

In [30]:
```
print("\nDescribe All Columns:")
print(df.describe(include='all'))
```

```
Describe All Columns:
                Age  Sex    BP Cholesterol     Na_to_K   Drug
count    200.000000  200   200         200  200.000000    200
unique          NaN    2     3           2         NaN      5
top             NaN    M  HIGH        HIGH         NaN  drugY
freq            NaN  104    77         103         NaN     91
mean      44.315000  NaN   NaN         NaN   16.084485    NaN
std       16.544315  NaN   NaN         NaN    7.223956    NaN
min       15.000000  NaN   NaN         NaN    6.269000    NaN
25%       31.000000  NaN   NaN         NaN   10.445500    NaN
50%       45.000000  NaN   NaN         NaN   13.936500    NaN
75%       58.000000  NaN   NaN         NaN   19.380000    NaN
max       74.000000  NaN   NaN         NaN   38.247000    NaN
```

In [31]:
```
# 1. Distribution of patients by gender - Bar Chart
plt.figure(figsize=(6, 4))
sns.countplot(x="Sex", data=df)
plt.title("Distribution of Patients by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Patients")
plt.show()
```



In [32]:
```
# 2. Which drug has the highest number of prescriptions?
drug_counts = df["Drug"].value_counts()

print("Number of prescriptions for each drug:")
print(drug_counts)

print("\nDrug with the highest number of prescriptions:")
print(drug_counts.idxmax())

print("Number of prescriptions:")
print(drug_counts.max())
```

```
Number of prescriptions for each drug:
Drug
drugY    91
drugX    54
drugA    23
drugC    16
drugB    16
Name: count, dtype: int64

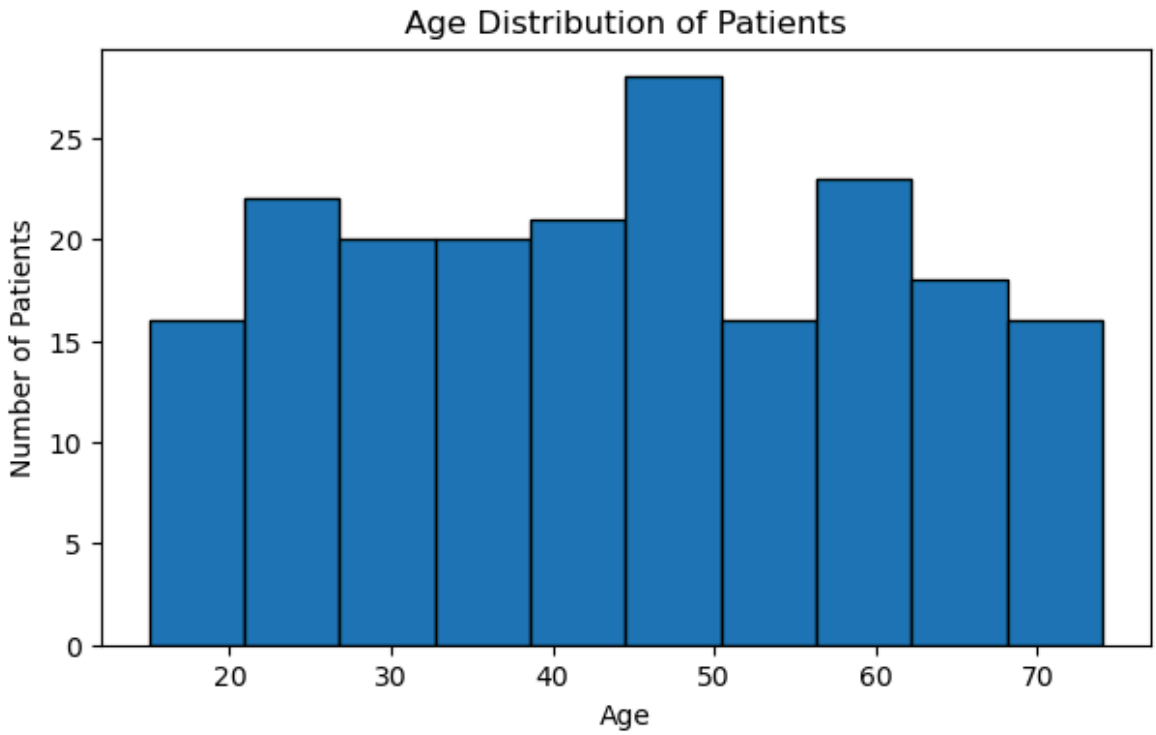
Drug with the highest number of prescriptions:
drugY
Number of prescriptions:
91
```

In [33]:
```
# 3. Proportion of each drug in the Drug200 dataset
drug_proportion = df["Drug"].value_counts(normalize=True) * 100

print("Proportion of each drug:")
print(drug_proportion)
```

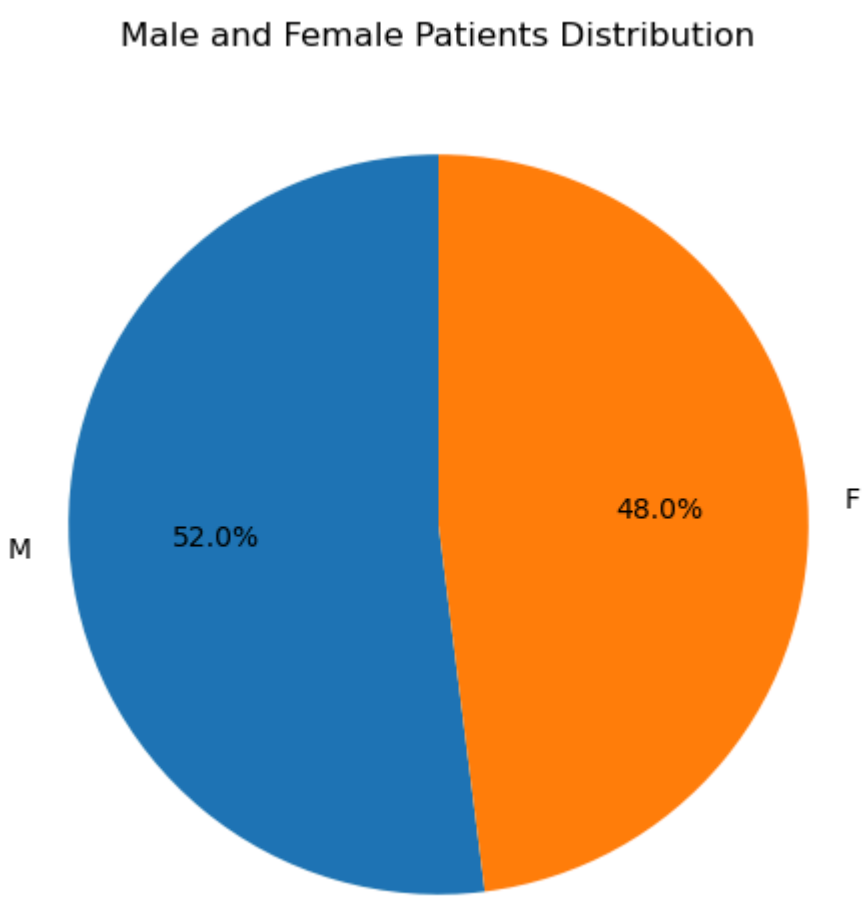
```
Proportion of each drug:
Drug
drugY    45.5
drugX    27.0
drugA    11.5
drugC     8.0
drugB     8.0
Name: proportion, dtype: float64
```

In [34]:
```
# 4. Age distribution of patients - Histogram
plt.figure(figsize=(7, 4))
plt.hist(df["Age"], bins=10, edgecolor="black")
plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Number of Patients")
plt.show()
```

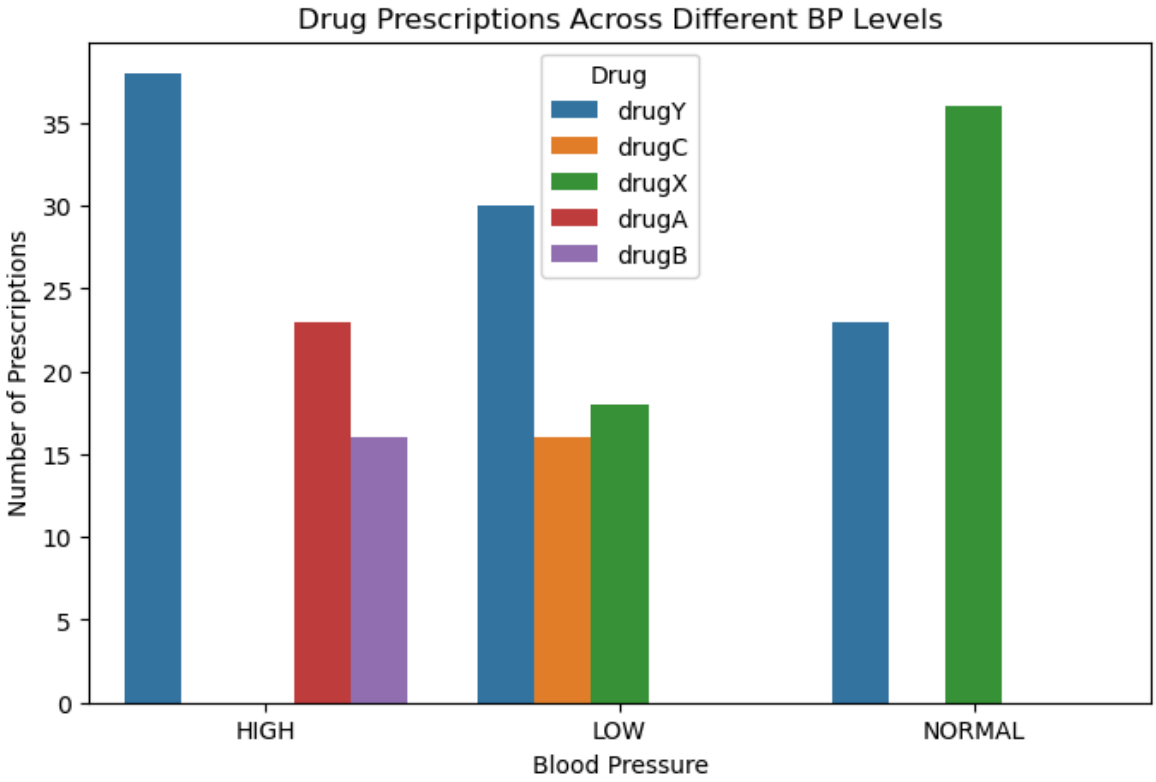


In [35]:
```
# 5. Male and Female patients - Pie Chart
sex_counts = df["Sex"].value_counts()

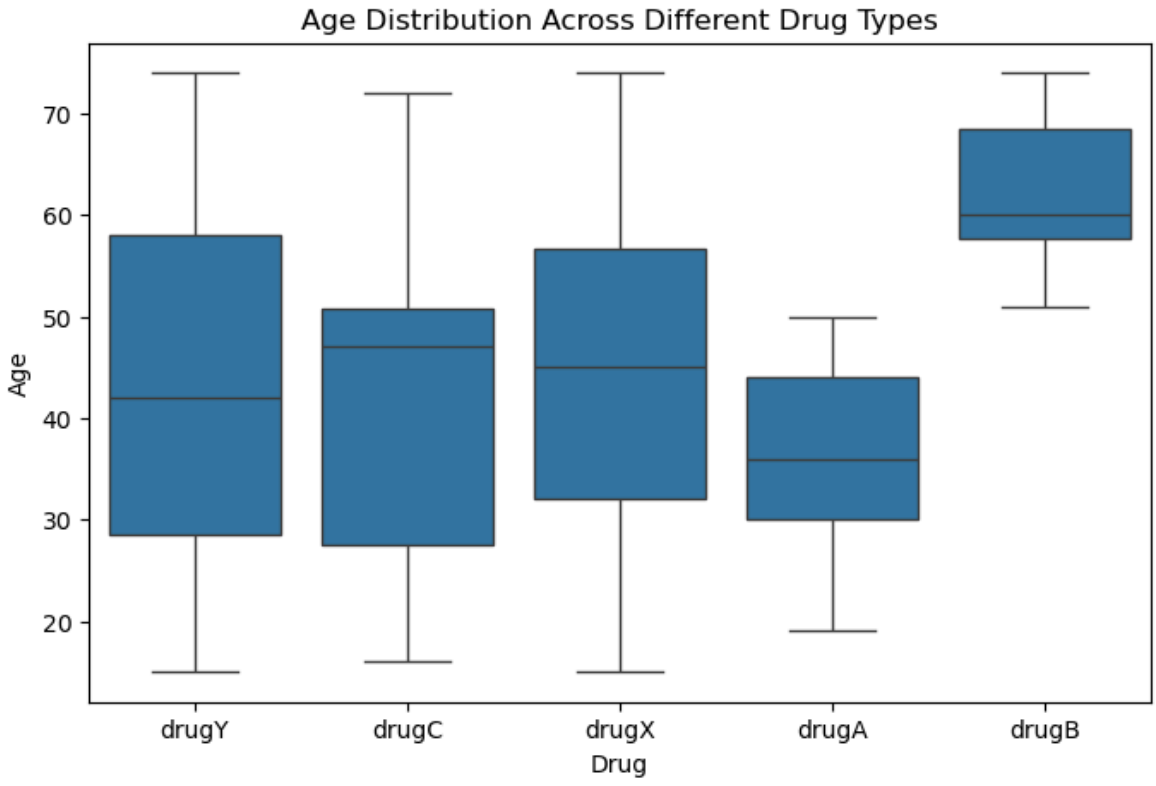
plt.figure(figsize=(6, 6))
plt.pie(
    sex_counts,
    labels=sex_counts.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Male and Female Patients Distribution")
plt.show()
```



In [36]:
```
# 6. Drug prescriptions across different BP levels - Count Plot
plt.figure(figsize=(8, 5))
sns.countplot(x="BP", hue="Drug", data=df)
plt.title("Drug Prescriptions Across Different BP Levels")
plt.xlabel("Blood Pressure")
plt.ylabel("Number of Prescriptions")
plt.show()
```

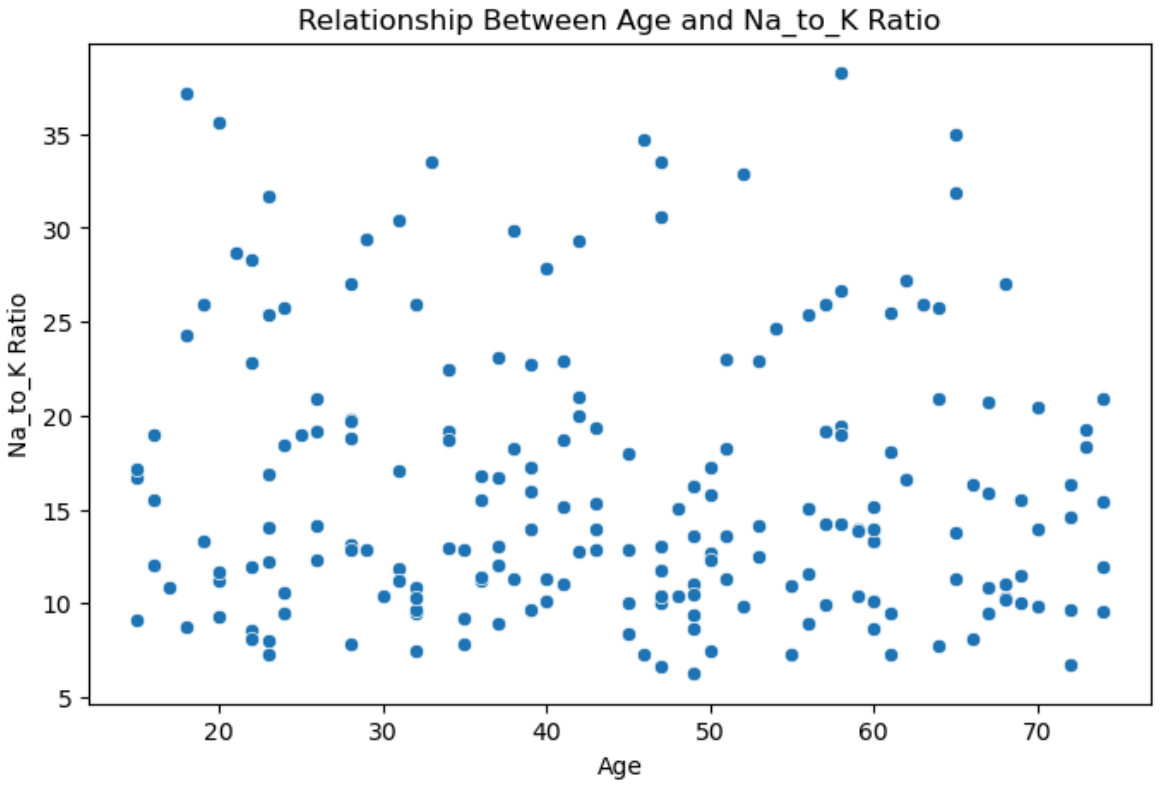


In [37]:
```
# 7. Age across different drug types - Box Plot
plt.figure(figsize=(8, 5))
sns.boxplot(x="Drug", y="Age", data=df)
plt.title("Age Distribution Across Different Drug Types")
plt.xlabel("Drug")
plt.ylabel("Age")
plt.show()
```



In [38]:
```
# 8. Relationship between Age and Na_to_K ratio - Scatter Plot
plt.figure(figsize=(8, 5))
sns.scatterplot(x="Age", y="Na_to_K", data=df)
```

```
plt.title("Relationship Between Age and Na_to_K Ratio")
plt.xlabel("Age")
plt.ylabel("Na_to_K Ratio")
plt.show()
```



In [ ]:

In [ ]: